

Observational Technique - Assignment 3

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1 Region of Interest

To determine region of interest (ROI), there are some specification:

1.1 Bias Frame

A bias frame is an image taken with zero exposure time and shutter closed. It records the readout electronics noise of the CCD detector, capturing the defects on the detector). To choose region of interest, we want region without detector defect or cosmic ray. Therefore, regions affected by hot pixels, cosmic rays, dead columns, as well as the edges and corners of the detector, should be avoided to ensure accurate bias level measurement. ROI shouldn't be too big (to avoid huge error). Here we use square ROI and use more than 30 frame in each arm to get better result (we use 41 files for each arm)

1.2 Flat field

A flat-field image is taken using a uniformly illuminated source (such as the inside of a telescope dome or the twilight sky) to correct for pixel-to-pixel sensitivity variations in the detector. Ideally, it has uniform brightness. Flat field has stripe, which resembles dark and bright region which caused by the way flat field taken. Here, we use the bright region (which is sensitive to light). Hence, we use rectangular size of ROI to capture larger region on the stripes. We use 36 files of flat fields in each arm.

2 Master bias frame

The master bias is created by stacking individual bias frames using the median value to remove outliers such as cosmic rays and hot pixels. This results in an image that captures the CCD's readout noise (including overscan region). Figure 1(a) and (b) show the master bias for each arm, while Figure 1(c) and (d) display a single bias frame, and Figure 1(e) and (f) show the result after subtracting the master bias. From the plot we can see that the overscan region are consistent in each frame. The subtraction will reduce pixel values to ideally zero, leaving only random noise. We use percentile [5,95] to show the cmap color variance on the plot. While the master bias reflects detector-specific features, subtraction still leaves some residual variance (e.g, cosmic rays) in individual frames (as expected).

3 Detector Linearity

Here we use flat-field images and select regions of interest (ROI) in the bright areas, which capture photons from a uniformly illuminated source. Figures 2(a) and (b) show the

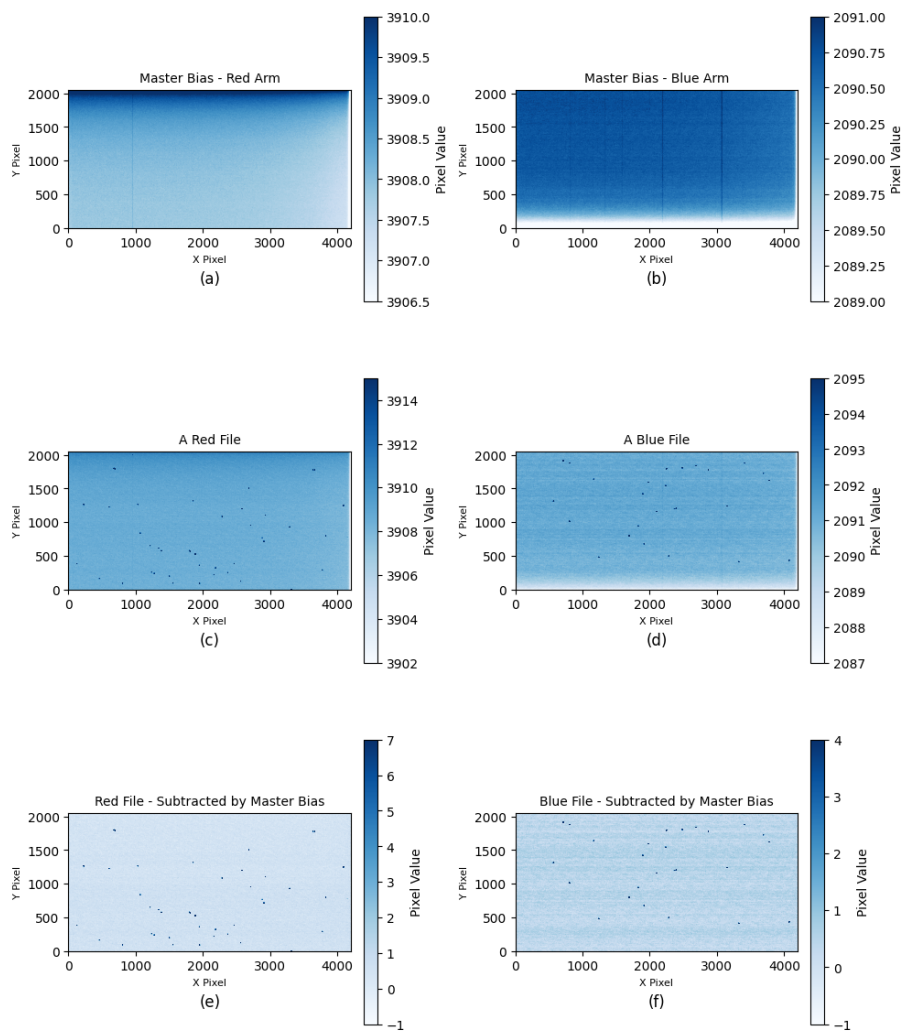


Figure 1: Master bias

chosen ROI on the flat-field data, after subtracted by the master bias. The corresponding detector linearity plots are shown in Figures 2(e) and (f).

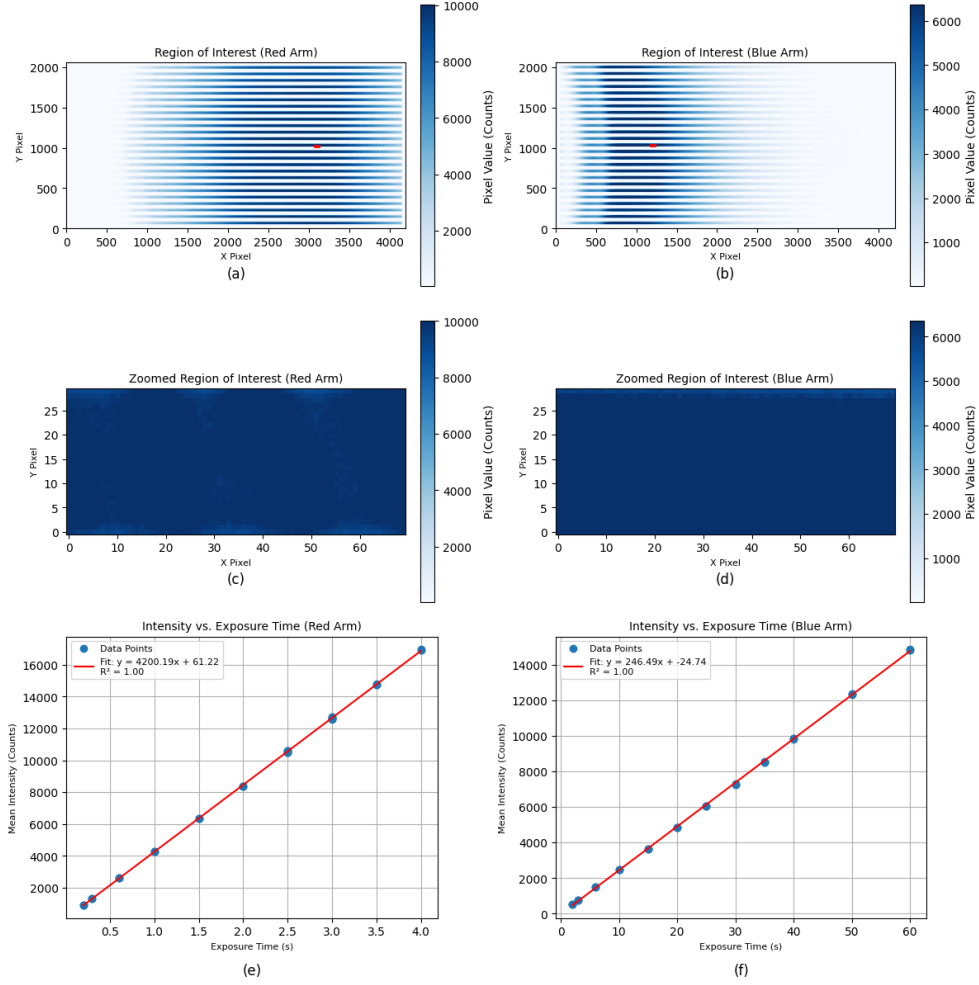


Figure 2: Detector linearity from selected ROI on flat-field data

In the bright region, the detector exhibits a linear response. The linearity shows $R^2 = 1$ which explains data points lie exactly on a straight line. In Figure 3, we test a different ROI, and the detector still shows a consistent linear response, as indicated by the measured values.

4 Initial Noise Estimate

The bias frames used in this analysis are as follows:

- Bias1 (Red Arm): OBK-709088-WiFeS-Red--UT20250314T230114-7.fits
- Bias2 (Red Arm): OBK-709088-WiFeS-Red--UT20250314T230211-9.fits
- Bias1 (Blue Arm): OBK-709088-WiFeS-Blue-UT20250314T231436-1.fits
- Bias2 (Blue Arm): OBK-709088-WiFeS-Blue-UT20250314T230211-9.fits

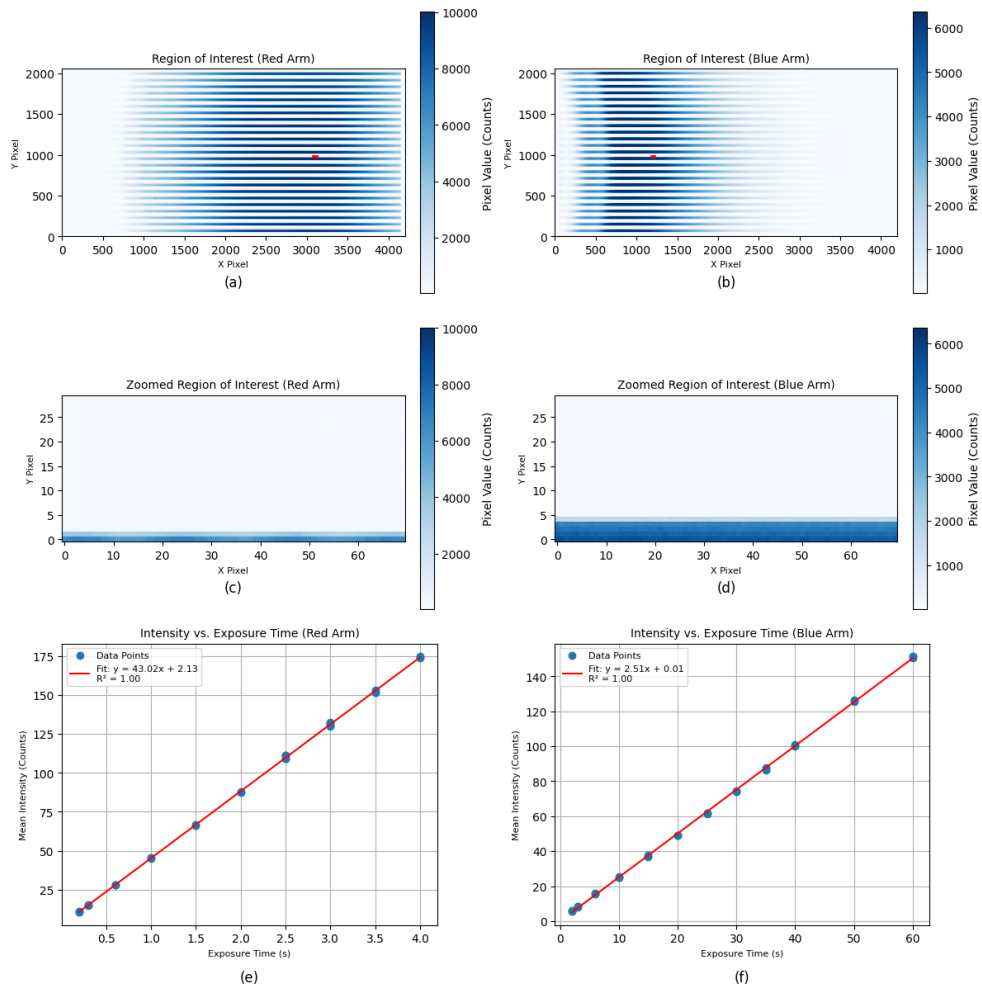


Figure 3: Detector linearity tested on a different ROI

4.1 Part (a)

In Part (a), we estimate the noise level from individual bias frames (see Figure 4). A region of interest (ROI), shown as red squares in Figure 4, was selected to avoid defects and ensure uniformity. Within this ROI, we calculated the standard deviation of pixel values and also generated a histogram of pixel intensity values, which we fit using a Gaussian function. The results for each arm are:

- **Bias 1 (Red Arm):**
 - Standard deviation: 4.02 ADU
 - Gaussian fit: $\mu = 3912.05$, $\sigma = 4.22$
- **Bias 1 (Blue Arm):**
 - Standard deviation: 2.38 ADU
 - Gaussian fit: $\mu = 2090.65$, $\sigma = 2.39$
- **Bias 2 (Red Arm):**
 - Standard deviation: 4.01 ADU
 - Gaussian fit: $\mu = 3908.61$, $\sigma = 4.43$
- **Bias 2 (Blue Arm):**
 - Standard deviation: 2.39 ADU
 - Gaussian fit: $\mu = 2089.01$, $\sigma = 2.35$

4.2 Part (b)

In Part (b), we estimate the noise by subtracting one bias frame from another (see Figure 5). Using the same ROI as before, we calculate the standard deviation of the resulting difference image and fit a Gaussian to the histogram of pixel values.

Ideally, subtracting two identical bias frames would result in a flat image at zero pixel. In reality, however, we observe a distribution due to pixel-level fluctuations, which can arise from cosmic rays, hot pixels, or other detector effects. Additionally, the histogram includes negative values, which are expected since the subtraction does not take the absolute value.

The measured results are:

- Standard deviation (Red Arm): 5.59 ADU
- Gaussian fit: $\mu = -3.35$, $\sigma = 5.72$
- Standard deviation (Blue Arm): 3.21 ADU
- Gaussian fit: $\mu = -1.68$, $\sigma = 3.33$

Theoretically, the standard deviation of the difference image should follow:

$$\sigma_{\text{diff}}^2 = \sigma_1^2 + \sigma_2^2 \tag{1}$$

Using the measured standard deviations from Part (a):

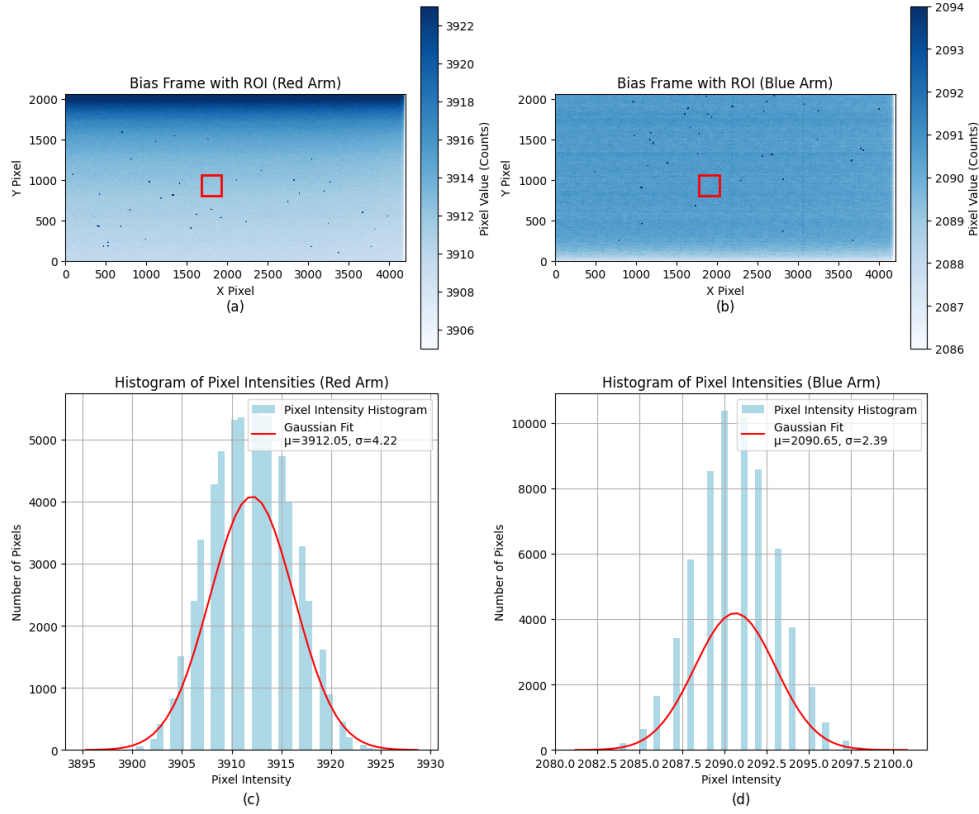


Figure 4: Noise estimate from a single bias frame (Bias 1) using the selected ROI.

- **Red Arm:** $\sigma_{\text{diff}}^2 = 4.01^2 + 4.02^2 = 32.16 \Rightarrow \sigma_{\text{diff}} \approx 5.68$ ADU
- **Blue Arm:** $\sigma_{\text{diff}}^2 = 2.38^2 + 2.39^2 = 11.34 \Rightarrow \sigma_{\text{diff}} \approx 3.37$ ADU

These predicted values are almost consistent (only slight offset) with the measured standard deviations from the difference image: 5.59 ADU (Red Arm) and 3.21 ADU (Blue Arm), confirming the reliability of our estimation.

5 Flat Field Noise

In this section, we use two flat-field frames with identical exposure times for each arm of the spectrograph. The files used are:

- Flat 1 (Red): OBK-709344-WiFeS-Red-UT20250315T000118-1.fits
- Flat 2 (Red): OBK-709344-WiFeS-Red-UT20250315T000426-4.fits
- Flat 1 (Blue): OBK-709440-WiFeS-Blue-UT20250315T000744-7.fits
- Flat 2 (Blue): OBK-709440-WiFeS-Blue-UT20250315T000605-6.fits

5.1 Part (a)

In Part (a), we estimate the standard deviation within a selected region of interest (ROI) (see Figure 6), similar to the method used in Question 4. The results for each flat frame are as follows:

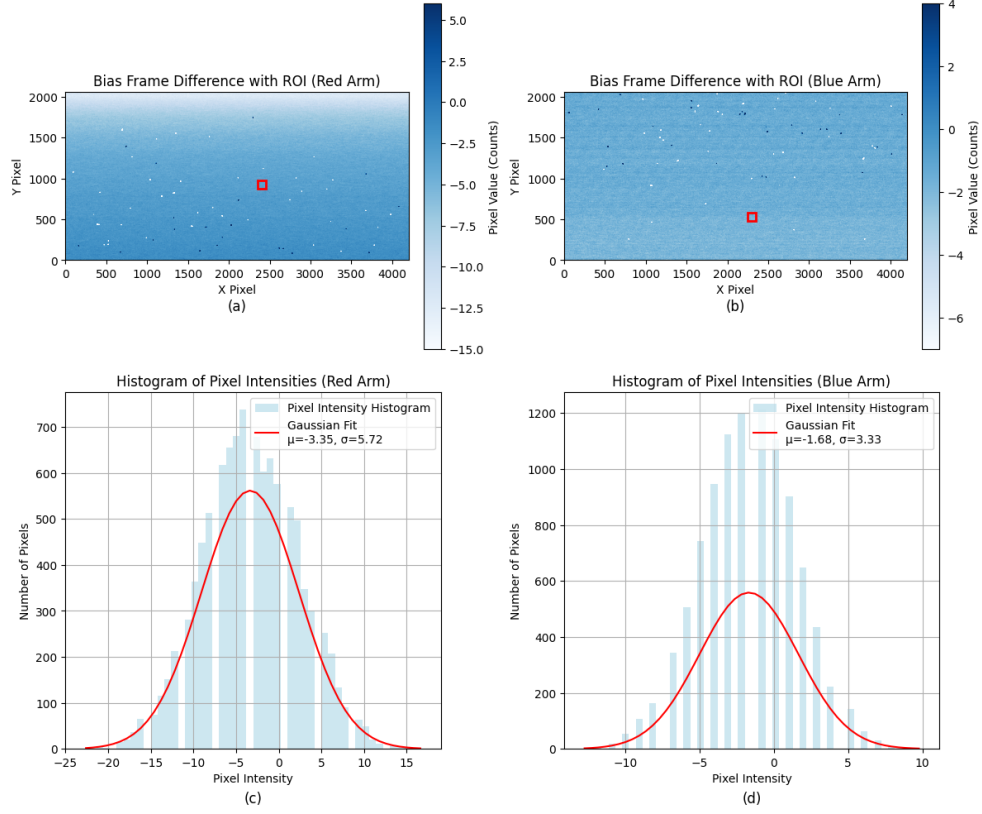


Figure 5: Noise estimate from the difference between two bias frames.

Flat 1:

- Standard deviation (Red Arm): 159.26 ADU
- Standard deviation (Blue Arm): 103.35 ADU
- Gaussian fit (Red Arm): $\mu = 10054.58$, $\sigma = 178.13$
- Gaussian fit (Blue Arm): $\mu = 8488.98$, $\sigma = 108.61$

Flat 2:

- Standard deviation (Red Arm): 164.85 ADU
- Standard deviation (Blue Arm): 105.52 ADU
- Gaussian fit (Red Arm): $\mu = 10045.40$, $\sigma = 182.14$
- Gaussian fit (Blue Arm): $\mu = 8449.85$, $\sigma = 103.74$

5.2 Part (b)

In Part (b), we estimate the noise by subtracting one flat-field frame from another (see Figure 7). The standard deviation and Gaussian fit parameters of the difference image are:

- Standard deviation (Red Arm): 122.28 ADU

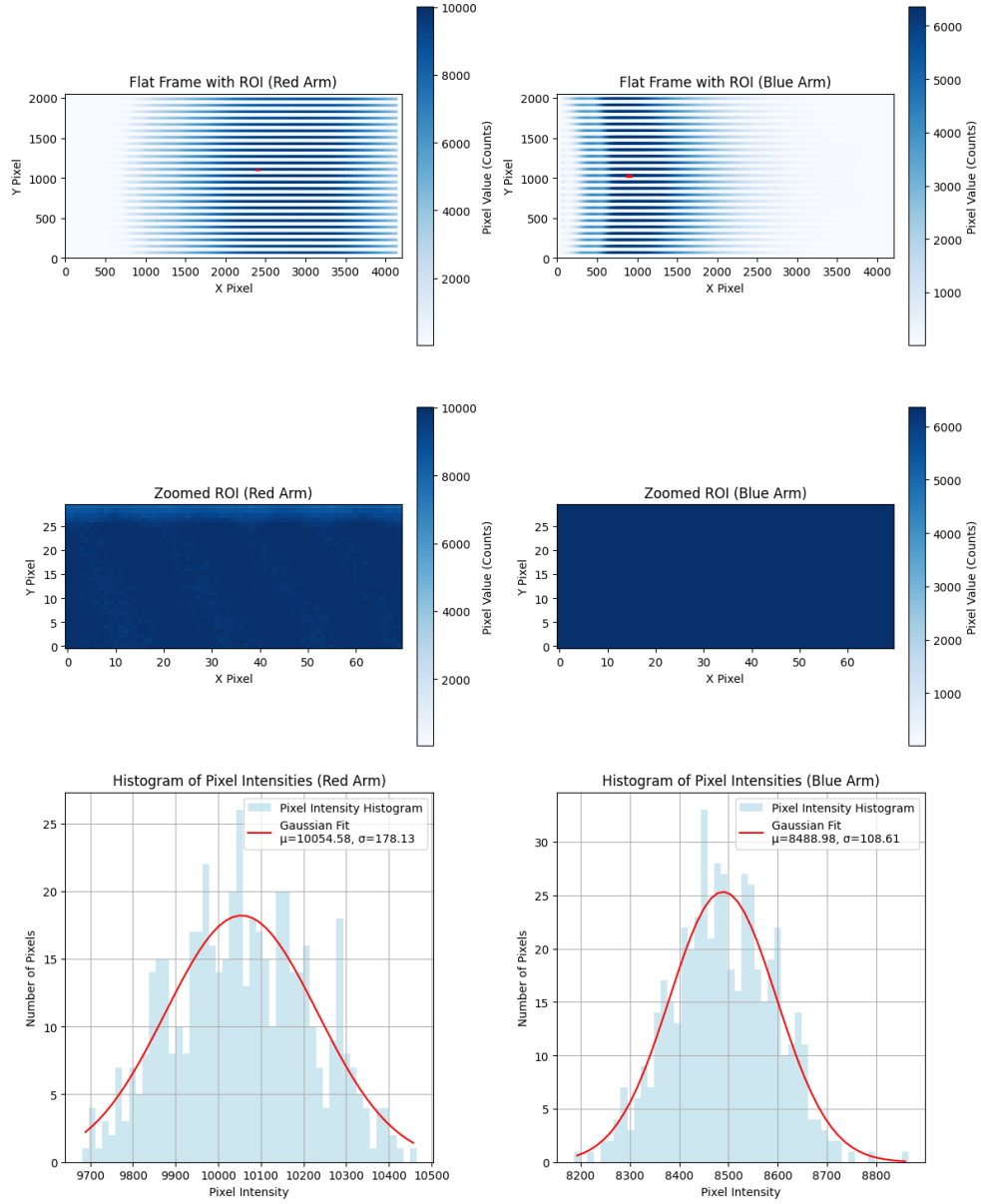


Figure 6: Noise estimate of Flat 1 using selected ROI.

- Gaussian fit (Red Arm): $\mu = 47.71$, $\sigma = 116.52$
- Standard deviation (Blue Arm): 97.12 ADU
- Gaussian fit (Blue Arm): $\mu = 27.87$, $\sigma = 86.08$

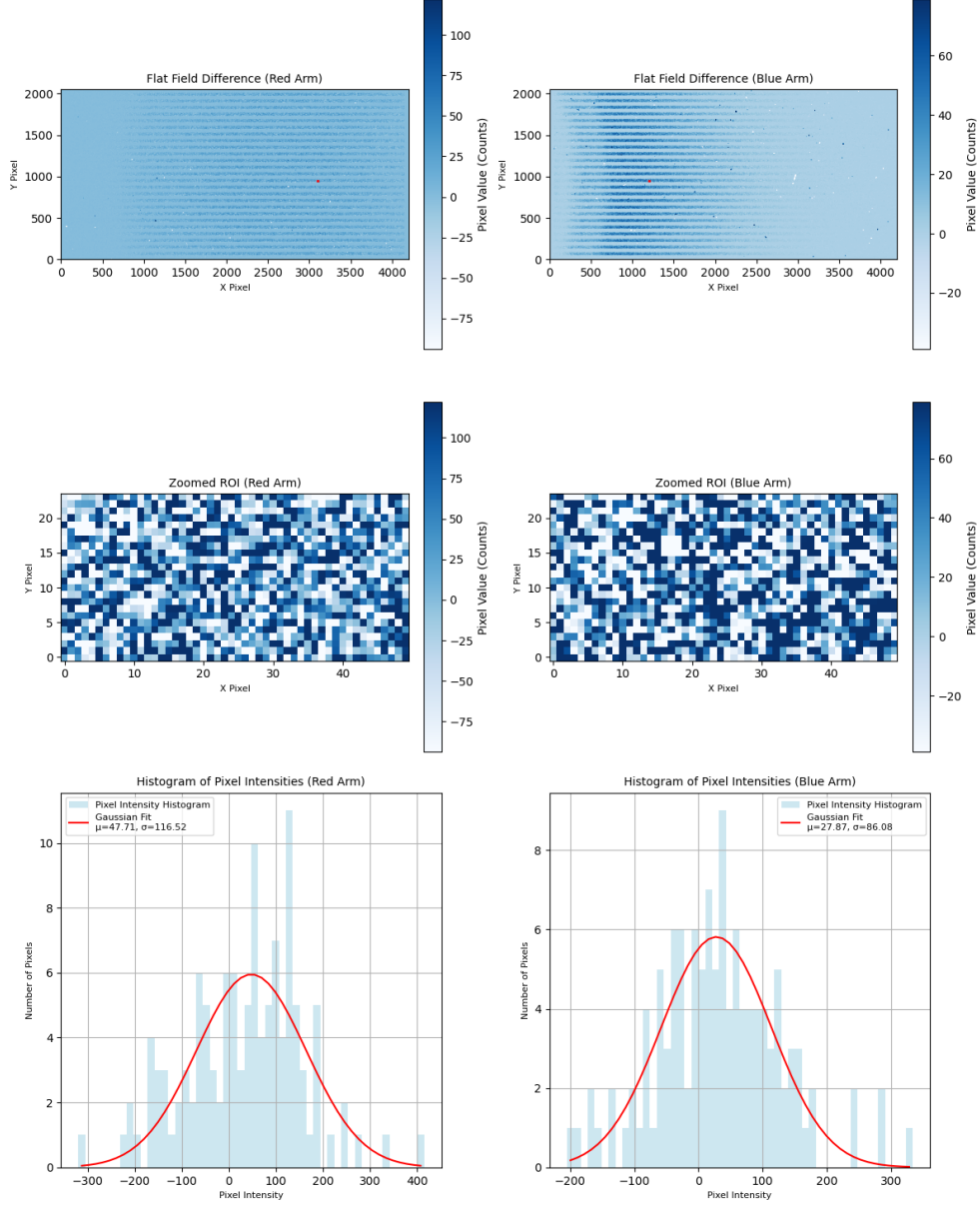


Figure 7: Noise estimate from the difference between two flat-field frames.

Using the theoretical noise propagation formula:

$$\sigma_{\text{diff}}^2 = \sigma_1^2 + \sigma_2^2 \quad (2)$$

We obtain the following estimates:

- **Red Arm:** $\sigma_{\text{diff}} = \sqrt{159.26^2 + 164.85^2} \approx 229.27$ ADU
- **Blue Arm:** $\sigma_{\text{diff}} = \sqrt{103.35^2 + 105.52^2} \approx 147.69$ ADU

In contrast to Question 4, the observed standard deviations in the difference image are *lower* than the theoretical predictions. As seen in Figure 7, after subtracting, the intensity varies and the intensity distribution is not centered at zero, suggesting a systematic offset between the two flat fields (one flat has a higher pixel value). This could arise from:

- Variations in the calibration lamp
- Imperfect bias subtraction (e.g., if the master bias is not representative of both flats)

Due to these systematic effects, pixel values in the two flats are not truly identical. Therefore, the standard deviations from Part (a) (which were calculated after bias subtraction) are more reliable for characterizing the flat-field noise.

6 Gain and Read Noise

To calculate the gain and read noise, we use pairs of flat-field frames with the same exposure time. The following equation is used to calculate the gain and read noise:

$$\sigma_{ADU}^2 = \frac{c_{ADU}}{g} + \left[\frac{1}{g^2} \cdot N_{rd}^2 \right] \quad (3)$$

The mean intensity is calculated using the average intensity of each pair, and the variance is calculated using the equation:

$$\sigma_{diff}^2 = \sigma_{sum}^2 = \sigma_A^2 + \sigma_B^2 = 2 \cdot \sigma_A^2 \quad (4)$$

The results are shown in Figure 8.

The region of interest (ROI) is chosen at the peak of the intensity where the intensity distribution is almost flat (we use a very thin ROI for this case), yielding a positive intercept. From the plot, we can calculate the gain using the following relation:

$$mx = \frac{c_{ADU}}{g} \quad \Rightarrow \quad g = \frac{1}{x}$$

For the red arm, $x = 0.75$, so the gain is $g = 1.33$. For the blue arm, $x = 0.62$, so the gain is $g = 1.61$. These values are close to 1 for both red and blue, indicating that the gain is correctly calculated.

The read noise is then calculated using:

$$c = \frac{1}{g^2} \cdot N_{rd}^2 \quad \Rightarrow \quad N_{rd} = \sqrt{c \cdot g^2}$$

For the red arm, with $c = 338.97$ and $g = 1.33$, the read noise is $N_{rd} = 24.48$ electrons. For the blue arm, with $c = 190.89$ and $g = 1.61$, the read noise is $N_{rd} = 22.25$ electrons.

Next, we calculate the read noise from the bias frame using the standard deviation from two bias frame (one frame subtracted by another one as in question 4b) (σ_{diff}):

$$\sigma_{read} = \frac{gain \cdot \sigma_{diff}}{\sqrt{2}} \text{ electrons}$$

For the red arm, with $\sigma_{diff} = 5.59$, $gain = 1.33$, the calculated read noise is $\sigma_{read} = 5.26$ electrons. For the blue arm, with $\sigma_{diff} = 3.21$, $gain = 1.61$, the calculated read noise is $\sigma_{read} = 3.66$ electrons.

This discrepancy arises because the read noise calculated from the "Variance vs Intensity Slope" method includes photon noise so it has higher noise, while the bias frame method only measures the read noise.

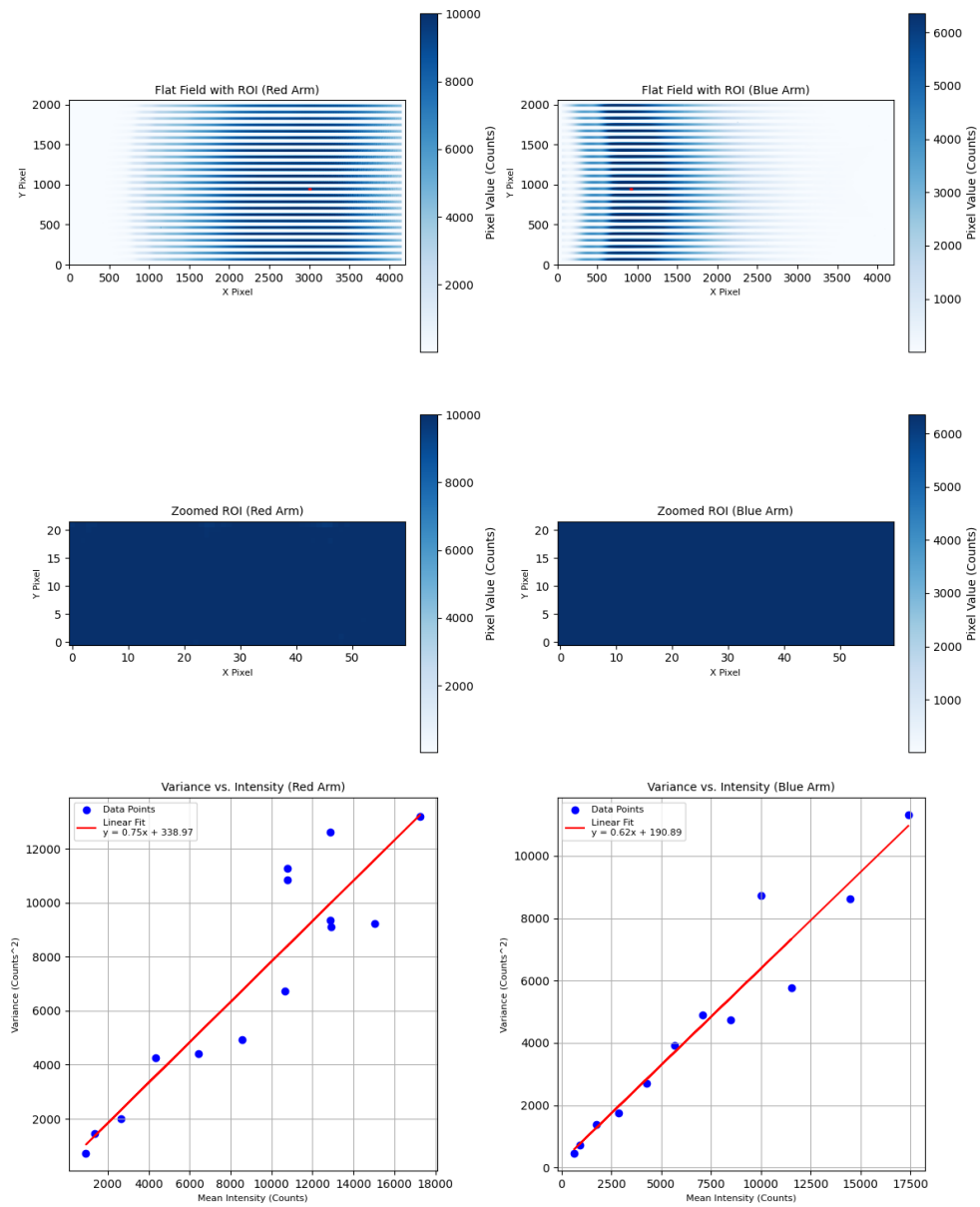


Figure 8: Intensity vs Variance.

7 Dark Current

The dark current is calculated using the following formula:

$$\text{mean signal} = \text{mean pixel value (ADU)} \times \text{gain} \quad (\text{e}^-) \quad (5)$$

The results are shown in Figure 9. Note that the dark current values have not been bias-subtracted.

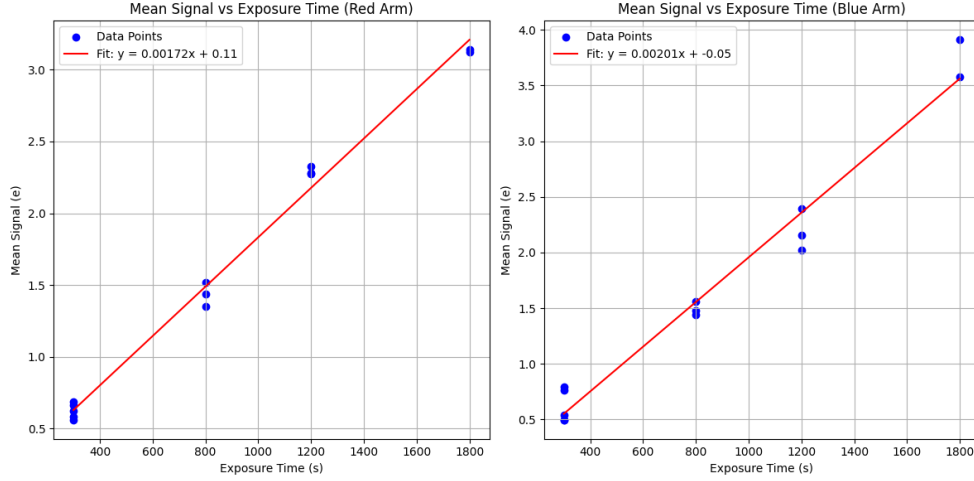


Figure 9: Mean signal as a function of exposure time for the red and blue arms.

Ideally, dark current should remain constant or increase linearly with time due to thermal effects. In our measurements, we observe an approximately linear trend. The dark current is calculated using the slope of the fitted line and converting to units of electrons per hour by multiplying by 3600. The results are:

- **Red Arm:**
 Slope: 0.00172 e/s
 Dark Current: 8.23 e⁻/hr
 R²: 0.99
- **Blue Arm:**
 Slope: 0.00201 e/s
 Dark Current: 11.63 e⁻/hr
 R²: 0.97

To determine the minimum exposure time required to avoid being read-noise limited, we use the condition where the noise from the dark current exceeds the read noise:

$$\sigma_{\text{dark}} > \sigma_{\text{read}}$$

where the noise from dark current is given by:

$$\sigma_{\text{dark}} = \sqrt{I_{\text{dark}} \cdot t}$$

Solving for t , the condition becomes:

$$t = \frac{\sigma_{\text{read}}^2}{I_{\text{dark}}}$$

Red Arm Calculation

- Read noise: $\sigma_{\text{read, red}} = 5.26 \text{ e}^-$
- Dark current: $D_{\text{red}} = 8.23 \text{ e}^-/\text{hr}$

$$t_{\text{red}} = \frac{(5.26)^2}{8.23} = \frac{27.65}{8.23} \approx 3.36 \text{ hr}$$

Blue Arm Calculation

- Read noise: $\sigma_{\text{read, blue}} = 3.21 \text{ e}^-$
- Dark current: $D_{\text{blue}} = 11.63 \text{ e}^-/\text{hr}$

$$t_{\text{blue}} = \frac{(3.21)^2}{11.63} = \frac{10.30}{11.63} \approx 0.89 \text{ hr}$$

These calculations indicate that exposures longer than approximately 3.36 hours (red arm) and 0.89 hours (blue arm) are required to ensure that observations are limited by dark current rather than by read noise.